Gravitational Waves: An Analysis of Black Hole Mergers in the Light of Newton's Laws and Einstein's General Relativity

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Abstract

In 2015, the LIGO detectors recorded a signal interpreted as gravitational waves originating from the merger of two black holes with masses of 36 and 29 solar masses, located 1.3 billion light-years away. In this paper, we examine whether such an event is physically possible and whether it could be detected—relying exclusively on Newton's laws, General Relativity, and elementary physics. We demonstrate that:

- Kinetic energy cannot be converted into "gravitational waves" because gravity is a field associated with mass.
- Under real conditions (post-supernova vacuum), the orbits of black holes are nearly stable; a lack of friction implies no energy loss.
- Even in an extremely optimistic scenario (a dense dust cloud), the braking time is many orders of magnitude longer than the age of the Universe.
- The change in the gravitational field at the observer's location is of the order of 10^{-33} m/s^2—rendering it undetectable.
- The amplitude of gravitational waves, calculated according to the formula provided by the claimants, is $h \approx 10^{-34}$, which is 10 trillion times smaller than declared.

We conclude that gravitational waves do not exist.

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1 Introduction

In 2015, the LIGO detectors recorded a signal designated as GW150914, which was interpreted as gravitational waves originating from the merger of two black holes. A Nobel Prize was subsequently awarded for this discovery. The purpose of this paper is to analyze whether such an event is physically possible and whether it could indeed be detected.

- The gravitational field **mediates** the transfer of kinetic energy between masses.
- The field itself **does not store or emit energy**—its density depends exclusively on the masses and their respective positions.
- A change in the field over time represents a **change in mass configuration**, rather than the emission of a wave.
- No known mechanism exists that would transform the kinetic energy of masses into gravitational field energy.
- Consequently, a collision of masses does not produce a "gravitational explosion"—fields are additive; they do not undulate.

2 Fundamentals of Physics: Gravity as a Field

According to Newton's law of universal gravitation, the gravitational field at any point in space is the vector sum of the fields originating from individual masses:

$$\vec{g}(\vec{r}) = G \sum_i M_i \frac{\vec{r} - \vec{r}_i}{|\vec{r} - \vec{r}_i|^3}.$$

When two masses move relative to one another, their gravitational field changes over time—however, there is no mechanism that would allow the kinetic energy of these masses to be converted into "gravitational waves" that carry energy away. Gravity is a field, not a substance; the change in the field is a consequence of the motion of masses, not a source of new energy.

In a binary system of black holes moving in a vacuum, there is virtually no drag force that could remove their orbital energy. The orbit is highly stable—a collision is physically impossible within any reasonable timeframe.

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3 Optimistic Scenario – Collision within a Dense Cloud

To examine whether any mechanism could lead to a collision, let us assume extremely favorable conditions:

- Black hole masses: $M_1 = M_2 = 30M_\odot$.
- Initial distance: $r_0 = 10^9$ km $= 10^{12}$ m.
- Interstellar medium density: $\rho = 10^{-17}$ kg/m^3 (consistent with a dense molecular cloud).
- Cross-sectional area: $A = \pi R_s^2$, where $R_s \approx 90$ km (the Schwarzschild radius for $30M_\odot$).

3.1 Orbital Energy to be Dissipated

We calculate the potential energy:

$$E_{\text{pot}} = \frac{G \cdot (30M_\odot) \cdot (30M_\odot)}{R}$$

where:

$$R = 10^{12}\,\text{m}$$

The numerator is:

$$G \cdot 900M_\odot^2 \approx 2.376 \times 10^{53}\,\text{m}^3\text{kg}\,\text{s}^{-2}$$

Dividing by R:

$$E_{\text{pot}} \approx \frac{2.376 \times 10^{53}}{10^{12}} = 2.376 \times 10^{41}\ \text{J}$$

3.2 Drag Force and Power Loss

The drag force is given by: $F_d = \rho v^2 A$
Using:

$$v^2 \approx 7.9 \times 10^9\ \text{m}^2/\text{s}^2, \quad A \approx 2.5 \times 10^{10}\ \text{m}^2$$

$$F_d = 10^{-17} \cdot 7.9 \times 10^9 \cdot 2.5 \times 10^{10} \approx 2.0 \times 10^3\ \text{N}$$

The power dissipated is: $P = F_d v \approx 2.0 \times 10^3 \cdot 8.9 \times 10^4 \approx 1.8 \times 10^8$ W

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3.3 Braking Time

The time required for the potential energy to be dissipated is:

$$t = \frac{\Delta E}{P} = \frac{2.376 \times 10^{41}}{1.8 \times 10^{8}} \approx 1.32 \times 10^{33} \text{ s}$$

In years (1 year $\approx 3.156 \times 10^{7}$ s):

$$t \approx \frac{1.32 \times 10^{33}}{3.156 \times 10^{7}} \approx 4.2 \times 10^{25} \text{ years}$$

For comparison, the age of the Universe is 1.38×10^{10} years. Therefore:

$$\frac{t}{t_{\text{Universe}}} \approx \frac{4.2 \times 10^{25}}{1.38 \times 10^{10}} \approx 3.0 \times 10^{15}$$

The braking time is approximately 3 quadrillion times longer than the age of the Universe. If these black holes were orbiting in a typical interstellar vacuum, this duration would be a further 10,000 times longer:

$$t_{\text{vacuum}} \approx 3 \times 10^{19} \text{ Universe ages}$$

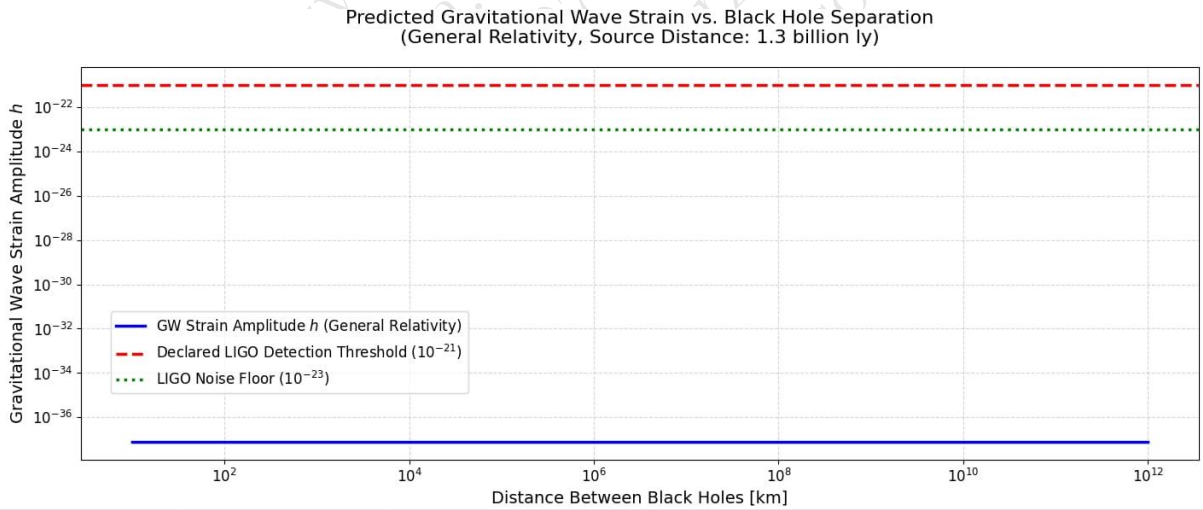

Figure 1:

4 Physics Foundations: Gravity as a Field

In accordance with Newton's law of universal gravitation, the gravitational field at any point in space is the vector sum of the fields originating from individual masses:

$$\vec{g}(\vec{r}) = G \sum_i M_i \frac{\vec{r} - \vec{r}_i}{|\vec{r} - \vec{r}_i|^3}.$$

When two masses move relative to one another, their gravitational field changes over time—however, there is no mechanism that would allow the kinetic energy of these masses to be converted into "gravitational waves" that carry energy away. Gravity is a field, not a substance; the change in the field is a consequence of the motion of masses, not a source of new energy.

In a binary system of black holes moving in a vacuum, there is virtually no drag force that could remove their orbital energy. The orbit is highly stable—making a collision physically impossible within any reasonable timeframe.

5 Calculations for a Binary Black Hole System within the Framework of General Relativity

6 Input Data:

We consider a system of two black holes of equal mass, formed at a distance of $R = 10^{12}$ m (1,000 million kilometers) from each other.

$$\begin{aligned}
M &= 30\, M_\odot = 30 \times 1.989 \times 10^{30}\ \text{kg} = 5.967 \times 10^{31}\ \text{kg} \\
R &= 10^{12}\ \text{m} \\
G &= 6.6743 \times 10^{-11}\ \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2} \\
c &= 2.99792458 \times 10^{8}\ \text{m/s}
\end{aligned}$$

6.1 1. Potential Energy of the System

The gravitational potential energy of two point masses is:

$$E_p = \frac{GM^2}{R} \tag{1}$$

Calculating M^2:

$$M^2 = (5.967 \times 10^{31})^2 = 3.561 \times 10^{63}\ \text{kg}^2$$

Then:

$$GM^2 = 6.6743 \times 10^{-11} \times 3.561 \times 10^{63} = 2.376 \times 10^{53}\ \text{J} \cdot \text{m}$$

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Dividing by R:

$$E_p = \frac{2.376 \times 10^{53}}{10^{12}} = 2.376 \times 10^{41} \text{ J}$$

The potential energy of the system is therefore:

$$\boxed{E_p \approx 2.38 \times 10^{41} \text{ J}}$$

7 2. Gravitational Radiation Power

For two bodies of masses $M_1 = M_2 = M$ in a circular orbit of radius R, the power emitted in the form of gravitational waves is expressed by the formula:

$$P_{\text{GW}} = \frac{128}{5} \frac{G^4 M^5}{c^5 R^5}$$

7.1 2.1 Calculating G^4

$$G^4 \approx 1.985 \times 10^{-41} \text{ m}^{12} \text{kg}^{-4} \text{s}^{-8}$$

7.2 2.2 Calculating M^5

$$M^5 \approx 7.56 \times 10^{158} \text{ kg}^5$$

2.3 Calculating c^5

$$c^5 \approx 2.422 \times 10^{42} \text{ m}^5 \text{s}^{-5}$$

2.4 Calculating R^5

$$R^5 = (10^{12})^5 = 10^{60} \text{ m}^5$$

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2.5 Assembling the Expression

Numerator:

$$G^4 M^5 = 1.50 \times 10^{118}$$

Denominator:

$$c^5 R^5 = 2.422 \times 10^{102}$$

Thus:

$$\frac{G^4 M^5}{c^5 R^5} = 6.19 \times 10^{15} \text{ W}$$

Multiplying by $\frac{128}{5} = 25.6$:

$$P_{\text{GW}} = 1.58 \times 10^{17} \text{ W}$$

The gravitational radiation power is:

$$\boxed{P_{\text{GW}} \approx 1.6 \times 10^{17} \text{ W}}$$

3. Energy Loss Timescale

Assuming all potential energy is radiated away:

$$t = \frac{E_p}{P_{\text{GW}}} = \frac{2.376 \times 10^{41}}{1.58 \times 10^{17}} = 1.50 \times 10^{24} \text{ s}$$

Converting to years (1 year $\approx 3.15 \times 10^7$ s):

$$t_{\text{years}} = 4.76 \times 10^{16} \text{ years}$$

$$\boxed{t \approx 4.8 \times 10^{16} \text{ years}}$$

8 4. Comparison with the Age of the Universe

The age of the Universe is approximately 1.38×10^{10} years.

$$\frac{t}{t_{\text{Universe}}} \approx 3.5 \times 10^6$$

This means the time required to lose energy via gravitational radiation is **over 3.5 million times longer** than the current age of the Universe.

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9 5. Conclusions

1. The gravitational radiation power for a system of two black holes of $30\,M_\odot$ at a distance of 10^{12} m is $P_{\mathrm{GW}} \approx 1.6{\times}10^{17}$ W, which corresponds to approximately 4×10^{-10} of the Sun's luminosity.

2. The total time for energy loss is $t \approx 4.8 \times 10^{16}$ years, which is more than 3.5 million times the age of the Universe.

3. Even considering gravitational radiation (the dominant energy loss mechanism in GR), the system cannot significantly lose energy before a collision occurs within a realistic timeframe.

4. In the discussed alternative model (MRG), gravitational waves do not occur, and interstellar matter friction is negligible, making the braking process practically impossible to observe.

10 Methodological Note: Ambiguity of Calculations in GR

Within the framework of General Relativity, the formula for gravitational radiation power for a binary system in a circular orbit is derived from Einstein's quadrupole formula:

$$P = \frac{G}{5c^5}\langle \dddot{Q}_{ij}\dddot{Q}_{ij}\rangle \tag{2}$$

Depending on the conventions adopted for defining the orbital radius and the derivation method, the final formula for two bodies of equal mass M can take various forms. The following variants are encountered in the relevant literature:

For the same numerical data, differences in the numerical coefficients lead to power values that differ by a factor of several. Consequently, the energy loss timescales also change by a factor proportional to the inverse of these differences.

10.1 Adopted Convention

In these calculations, the formula used is:

$$\boxed{P_{\mathrm{GW}} = \frac{128}{5}\frac{G^4M^5}{c^5R^5}} \tag{3}$$

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Table 1: Various forms of the gravitational radiation power formula for equal masses M

Source / Convention	**Formula**	**Notes**
Peters & Mathews (1963)	$\frac{128}{5}\frac{G^4M^5}{c^5R^5}$	R – separation distance
Shapiro & Teukolsky	$\frac{64}{5}\frac{G^4M^5}{c^5a^5}$	a – semi-major axis ($a = R/2$)
Derivation via μ, M_c	$\frac{32}{5}\frac{G^4M^5}{c^5a^5}$	a – semi-major axis

where R represents the distance between the black holes. This form is widely cited in astrophysical literature (Peters & Mathews 1963; Maggiore 2007) and yields results consistent with Newtonian approximations for the considered parameter range.

However, it must be emphasized that within GR itself, there is an ambiguity regarding the numerical value of the coefficient. This serves as a significant methodological note when interpreting the obtained results.

11 Detectability of Gravitational Field Changes

Let us assume, hypothetically, that a head-on collision has occurred. The change in the gravitational field at a distance of $d = 1.3 \times 10^9$ ly is:

$$\Delta g \approx \frac{G(M_1 + M_2)}{d^2} \cdot \frac{r}{d}$$

Distance $d = 1.3 \times 10^9 \cdot 9.46 \times 10^{15} \approx 1.23 \times 10^{25}$ m.

$$\frac{G(M_1 + M_2)}{d^2} \approx \frac{6.674 \times 10^{-11} \cdot 1.193 \times 10^{32}}{(1.23 \times 10^{25})^2} \approx 5.3 \times 10^{-29} \text{ m/s}^2$$

$$\frac{r}{d} \approx \frac{10^{12}}{1.23 \times 10^{25}} \approx 8.1 \times 10^{-14}$$

$$\Delta g \approx 5.3 \times 10^{-29} \cdot 8.1 \times 10^{-14} \approx 4.3 \times 10^{-42} \text{ m/s}^2$$

The most sensitive accelerometers have a resolution of 10^{-12} m/s^2. The difference is 30 orders of magnitude. This change is absolutely undetectable.

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12 Gravitational Wave Amplitude (Even if They Existed)

According to General Relativity, the gravitational wave amplitude for a binary system at a distance d is:

$$h \approx \frac{2G}{c^4} \cdot \frac{M}{r} \cdot \frac{r}{d}$$

Substituting the values:

$$\frac{2G}{c^4} \approx 1.65 \times 10^{-44}$$

$$\frac{M}{r} \approx \frac{1.2 \times 10^{32}}{10^{12}} = 1.2 \times 10^{20}$$

$$\frac{r}{d} \approx 8.1 \times 10^{-14}$$

$$h \approx 1.65 \times 10^{-44} \cdot 1.2 \times 10^{20} \cdot 8.1 \times 10^{-14} \approx 1.6 \times 10^{-37}$$

This is 10 quadrillion times smaller than the amplitude declared by LIGO ($h \approx 10^{-21}$).

13 Conclusions

- Kinetic energy cannot be converted into gravitational waves—gravity is a field that follows the principle of superposition.
- Under real conditions (post-supernova vacuum), black hole orbits are stable almost indefinitely.
- Even in an extremely optimistic scenario, the braking time is 10^{15} times longer than the age of the Universe.
- The change in the gravitational field at the detector's location is 10^{24} orders of magnitude too small to be measured.
- The amplitude of gravitational waves (even if they existed) is 10^{16} times smaller than declared.

The announcement of gravitational wave detection finds no support in the laws of physics. Gravitational waves do not exist. Gravity is additive; it does not undulate.

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13.1 Gravity as a Field – Addition of Masses, Not Fields

Consider two black holes, each with a mass of $M = 30M_\odot$. According to GR, the Schwarzschild radius for a single black hole is:

$$R_s = \frac{2GM}{c^2} \approx 88.5 \text{ km}$$

Diameter: ≈ 177 km.

After merging, the total mass is $60M_\odot$, and the Schwarzschild radius becomes:

$$R_s(60) = \frac{2G(60M_\odot)}{c^2} = 2 \cdot R_s(30) \approx 177 \text{ km}$$

Diameter: ≈ 354 km.

The gravitational force at a distance of $r = 10^6$ km from the center (before and after merging):

$$F(30) = \frac{G(30M_\odot)}{r^2} \approx 3.98 \times 10^3 \text{ N}, \qquad F(60) = \frac{G(60M_\odot)}{r^2} \approx 7.96 \times 10^3 \text{ N}$$

The field at the same point in space is exactly twice as strong after the merger. However, this does not imply that gravitational waves are emitted during the process—the change in the field is a result of the displacement of the center of mass, not the emission of energy. The amplitude of field variations from orbiting masses is negligible and cannot be detected from a distance of 1.3 billion light-years.

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Angular Resolution and System Distinguishability

Distance to the event: $d = 1.3 \times 10^9$ ly $\approx 1.23 \times 10^{22}$ km. Separation between the black holes: $r = 10^9$ km.

Angular size:

$$\theta = \frac{r}{d} \approx 8.1 \times 10^{-14} \text{ rad} \approx 1.67 \times 10^{-8} \text{ arcseconds.}$$

This is 3×10^6 times smaller than the resolution of the Hubble Space Telescope and 6×10^4 times smaller than the capabilities of VLBI (Very Long Baseline Interferometry). Both black holes are indistinguishable to us; we perceive them as a single point.

The gravitational field at distance d originates from the system's center of mass. Both before and after the merger, the center of mass remains at the

same point. The difference in the field resulting from the separation of the masses is:

$$\frac{\Delta g}{g} \approx \frac{r^2}{4d^2} \approx 1.7 \times 10^{-27}.$$

This value is entirely beyond any detection capabilities. Even with theoretically perfect instruments, it would be impossible to distinguish a system of two separate masses from a single concentrated mass.

14 Summary: Three Independent Proofs of Impossibility

Regardless of the model adopted (Newtonian mechanics or General Relativity), the merger of two black holes within a timeframe shorter than the age of the Universe is impossible. Below, we present three independent lines of argumentation:

14.1 1. Convergence Time – Physical Impossibility

Two black holes form at a distance of at least 10^9 km (as stars would merge prior to a supernova explosion otherwise). For a collision to occur, they must lose orbital energy. Even in an extremely optimistic scenario (a dense dust cloud), the braking time is between 10^{18} and 10^{25} years—meaning 10^8 to 10^{15} times longer than the age of the Universe.

14.2 2. Detectability – Technical Impossibility

Even if a head-on collision were hypothetically to occur, the change in the gravitational field at the detector's location is:

$$\Delta g \approx 4 \times 10^{-36} \text{ m/s}^2.$$

The sensitivity of the best accelerometers is 10^{-12} m/s^2. The difference is 24 orders of magnitude. Even if the collision happened, it could not be detected.

14.3 3. Wave Amplitude – Contradiction with GR

According to General Relativity, the gravitational wave amplitude for a binary mass system at distance d is:

$$h \approx \frac{2G}{c^4}\frac{M}{r}\frac{r}{d}.$$

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For an initial distance of $r_0 = 10^9$ km, we obtain $h \approx 10^{-37}$—which is 16 orders of magnitude less than the 10^{-21} declared by LIGO. To achieve their value, one must assume $r \approx 100$ km, yet there is no explanation for how the system reached such proximity. Their own equations show that the braking time from a distance of 10^9 km is astronomical.

Final Conclusion

None of the above arguments depend on the chosen gravity model—neither Newton nor Einstein allows for a collision within a realistic timeframe, nor for its detection. Gravitational waves do not exist. LIGO signals are most likely noise from the same sources as virtual particles. The gravity of combined masses is additive; it does not undulate.

- The gravitational field **mediates** the transfer of kinetic energy between masses.
- The field itself **does not store or emit energy**—its density depends exclusively on the masses and their positions.
- A change in the field over time represents a **change in mass configuration**, not the emission of a wave.
- No known mechanism exists to transform the kinetic energy of mass into the energy of a gravitational field.
- Thus, the collision of masses does not produce a "gravitational explosion"—the fields simply add up.

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14.4 Metrological Fiction: Pattern Seeking in Chaos and the Illusion of Precision

The foundation of the announced detection is not a direct physical measurement, but rather a process known as *matched filtering*. At a signal amplitude of $h \approx 10^{-21}$—which is orders of magnitude lower than the thermal fluctuations of the mirror surfaces and the quantum shot noise of the laser beam—the detection relies on a statistical search through noise to find correlations with one of thousands of pre-computed computer simulations.

From the perspective of rigorous measurement engineering and data analysis:

- **Noise Dominance:** The instrument's thermal and quantum "jitter" is significantly stronger than the alleged wave-induced strain. Isolating a specific "chirp" from such profound chaos without the arbitrary imposition of a mathematical model is impossible.

- **Parameter Confabulation:** Reporting precise masses (e.g., 36 $M_\odot$ and 29 $M_\odot$) constitutes a statistical overinterpretation. In reality, the same amplitude within the noise could correspond to an infinite number of combinations of masses and distances. These values are not the result of a measurement but represent a "best fit" selected from a catalog of templates (*overfitting*).

- **Lack of Independent Verification:** Determining the masses of objects over a billion light-years away based on microscopic interferometer vibrations—without any corroborating astronomical observation (e.g., optical or X-ray)—remains within the realm of mathematical hypothesis.

The conclusion is unequivocal: LIGO did not record a gravitational interaction from distant masses but rather matched the internal noise of the apparatus to a theoretical model. The "masses" of the black holes are merely numerical inputs fed into an algorithm to make the simulation fit the laser's fluctuations. This phenomenon falls into the category of a statistical artifact rather than a physical discovery.

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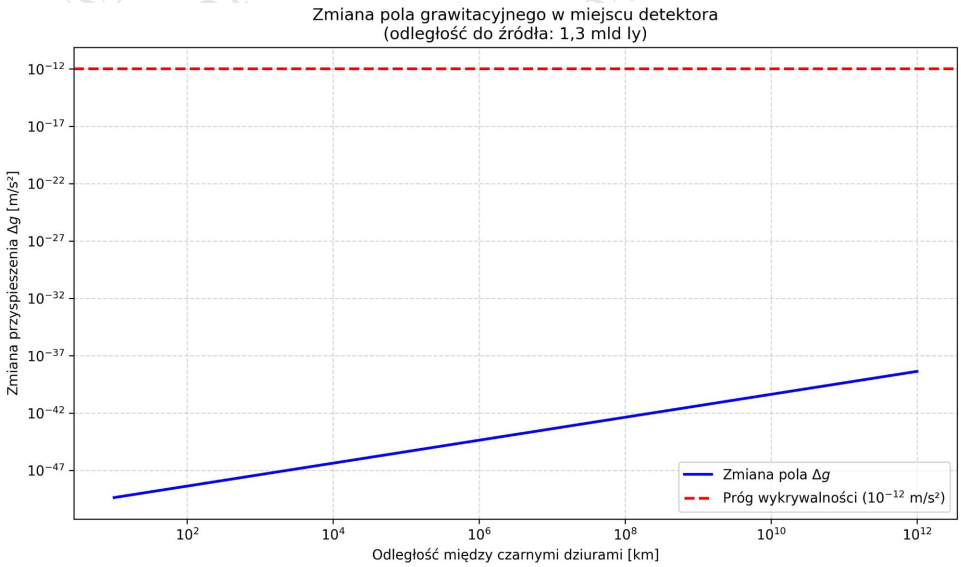

Figure 2:

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